

TRACK

Track 1, open-ended vault

FLAVOUR

Loaded: XLS-65, XLS-66,
TokenEscrow

ENVIRONMENT

Custom Hackathon Devnet, `rippled 3.4.0-rc1` (hackathon
branch), `LendingProtocolV1_1` and `fixCleanup3_4_0` enabled

LIBRARY AND VERSION

`xrpl@4.6.0`

The 143 transactions cited across the deliverables (this report, the README and the raw log) are validated on the ledger, type and result code included (`_probe-audit-hashes.mjs`). Documentation claims are checked against XLS-65, XLS-66 and `xrpl.org`, library claims against the `rippled` and `xrpl.js` source. Hashes link to the explorer.

#	Finding	Category	Severity
1	<code>fixCleanup3_4_0</code> is enabled and missing from <code>xrpl.org</code>	client libraries	HIGH
2	A late <code>LoanPay</code> without <code>tfLoanLatePayment</code> returns <code>tecEXPIRED</code>	documentation/tutorials	HIGH
3	First-loss capital covered 0.5% of a defaulted loan	UX	HIGH
4	A share <code>Payment</code> to a new holder returns <code>tecNO_AUTH</code>	documentation/tutorials	MEDIUM
5	Vault shares carry <code>lsfMPTCanTrade</code> , but <code>OfferCreate</code> rejects them	documentation/tutorials	MEDIUM
6	No read command for loans or brokers	missing primitive	MEDIUM
7	<code>xrpl.org</code> omits the sole-holder exception, and zero-valued fields are absent	documentation/tutorials	MEDIUM
8	One result code covers causes that need different actions	UX	MEDIUM
9	No flag closes a vault to deposits	missing primitive	LOW

High: blocks a minimum bar step, or misstates a depositor's risk.**Medium:** costs time or a wrong design, with a workaround.**Low:** cheap to work around, or cosmetic.

The brief's questions

Vault, broker, loan	Clear. Only the vault owner can submit <code>LoanBrokerSet</code> . The drawdown has no transaction: <code>LoanSet</code> moves the principal (<code>D19CD59F...</code>).
First-loss parameters	Not as named (3).
Multi-party <code>LoanSet</code>	Broker signs, borrower countersigns and submits. The <code>xrpl@4.6.0</code> helper fails here (1).
SDK or raw JSON	All 15 XLS-65/66 types are typed. No raw JSON.
Position and yield	Computed client-side from five calls (6).
Docs and explorer	The docs differ in 1, 2, 4, 5, 7. The explorer hides a flag name (2) and <code>Vault</code> fields (7).

1 `fixCleanup3_4_0` is enabled and missing from `xrpl.org` HIGH client libraries · `xrpl@4.6.0`

The amendment, new in `rippled 3.4.0-rc1`, is active on the devnet and absent from *Known Amendments*. It changes two lending behaviours, and the tutorial reflects neither.

Impairment. XLS-66, section 3.10.4.2, condition 9 rejects `tfLoanImpair` while `currentTime <= NextPaymentDueDate`. The tutorial *Manage a Loan* still says a broker can impair “before a payment due date passes”.

Counterparty signature. Since `rippled` PR #8162, `CounterpartySignature` uses the CPT prefix under the amendment, the previous prefix without it. `signLoanSetByCounterparty` has no option: `xrpl@4.6.0` always signs with the previous prefix, `xrpl@5.2.0` (`xrpl.js` PR #3462) always with CPT, which `rippled` rejects on networks without the amendment.

Repro steps

- `LoanSet` with `PaymentInterval 60` and `GracePeriod 60`. `LoanManage tfLoanImpair` returns `tecT00_S00N 31 s` and `13 s` before `NextPaymentDueDate`, and `tesSUCCESS 5 s` after, by the script's clock. The ledger closes the accepted transaction 8 s after the due date.
- The broker signs a `LoanSet`, the borrower applies `signLoanSetByCounterparty` and submits: `Counterparty: Invalid signature`, a local check with no hash.
- The same blob signed over `encodeForSigningCounterparty` (`raw-submit.mjs`) returns `tesSUCCESS`.

Transactions impairment `5486020B...`, countersigned `LoanSet 23631C36...`

Proposed fix List `fixCleanup3_4_0` on *Known Amendments* with both lending changes. Update *Manage a Loan*: impairment is accepted once `NextPaymentDueDate` has passed. In `xrpl.js`, pick the counterparty prefix from the network's `fixCleanup3_4_0` status, as `rippled` does.

2 A late `LoanPay` without `tfLoanLatePayment` returns `tecEXPIRED` HIGH documentation/tutorials · `xrpl@4.6.0`

Once `NextPaymentDueDate` has passed, every `LoanPay` without the flag fails, even within the grace period: the instalment, the instalment with `LatePaymentFee`, the full balance, and `tfLoanFullPayment`. XLS-66, section 3.11.4.2, condition 11 defines this failure; the `xrpl.org` `LoanPay` reference lists eight error codes, not this one. The explorer shows the flag as `0x00040000`, and no `Loan` field holds the accrued late interest.

Repro steps

- `LoanSet` with `PaymentInterval 90`, `GracePeriod 60`, `PaymentTotal 4` (`_probe-pay-states.mjs`).

- 11 s after the due date by ledger close time, `LoanPay` of `ceil(PeriodicPayment) + LoanServiceFee + LatePaymentFee` returns `tecEXPIRED` with `Flags: 0`, and `tesSUCCESS` with `Flags: 262144`.
- With `LateInterestRate 30%` ([_probe-h2-race.mjs](#)), the same amount returns `tecINSUFFICIENT_PAYMENT`. Sending 2,765,001 drops succeeds and takes 765,003.

Transactions no flag [96C38704...](#), flag [EFAD383F...](#), late interest [9A758938...](#)

Proposed fix Add `tecEXPIRED` to the `LoanPay` error table, with `tfLoanLatePayment` as the remedy. Name the flag in the explorer. Expose the amount due now, late interest included, in a `Loan` field or a read command.

Fail - TecEXPIRED

Status
 This transaction failed with a status code of `tecEXPIRED`, and validated in ledger `67844` on September 12, 2026 at 2:33:42 PM UTC

Success

Flags
`0x00040000`

Explorer. Same loan and amount, without the flag, then with it.

3 First-loss capital covered 0.5% of a defaulted loan

HIGH UX · xrp1@4.6.0

Two brokers, both with `CoverRateMinimum 10%` and `CoverRateLiquidation 5%`:

Vault	Loan	Cover	Taken	Loss	Share value
200 XRP	50	50	0.25	49.75	1.000 → 0.751
5 XRP	4	1	0.02	3.98	1.000 → 0.204

The cover taken is $\text{debt} \times 10\% \times 5\%$, to the drop. This is the documented formula: `CoverRateLiquidation` applies to the minimum cover, not to the defaulted debt, and the [xrpl.org](#) example covers 1% of the debt. After the default `DebtTotal` is zero, so is the cover floor, and the broker withdrew the rest. Second vault, UTC:

14:23:50	<code>VaultWithdraw: tecINSUFFICIENT_FUNDS</code>	D91BA47E...
14:27:01	<code>tfLoanDefault: 0.02 XRP from the cover</code>	923F7D61...
14:27:11	<code>VaultWithdraw: 1.02 XRP</code>	3338BA62...
14:27:22	<code>LoanBrokerCoverWithdraw: 0.98 XRP</code>	68DD97D9...

Repro steps

- [h1-default.mjs](#), after [setup-accounts.mjs](#): a 200 XRP vault, 50 XRP of cover, one 50 XRP loan, defaulted after `GracePeriod`.
- Compare `CoverAvailable` and `AssetsTotal` before and after the default.
- `LoanBrokerCoverWithdraw` of the remaining cover returns `tesSUCCESS`.

Transactions first vault default [91F458E3...](#)

Proposed fix Next to the formula, state the share of a defaulted debt the cover absorbs: $\text{CoverRateMinimum} \times \text{CoverRateLiquidation}$, 0.5% at 10% and 5%. Hold `LoanBrokerCoverWithdraw` for a set delay after a default, or state that no delay exists.

4 A share Payment to a new holder returns tecNO_AUTH

MEDIUM documentation/tutorials · xrp1@4.6.0

The Single Asset Vault page, *Can a Depositor Transfer Shares to Another Account?*, lists five failure causes, then: “Otherwise, a new MPT entry is created for their account.” Without an `MPToken` for the share issuance, the payee gets `tecNO_AUTH`, a cause the page omits, until it submits `MPTokenAuthorize`.

Repro steps

- Public XRP vault, no freeze. A share `Payment` to an account without `MPToken`: `tecNO_AUTH` ([_probe-shares2.mjs](#)).
- The payee submits `MPTokenAuthorize`, and the same `Payment` returns `tesSUCCESS` ([_probe-shares3.mjs](#)).

Transactions refused [789AAF7E...](#), opt-in [3AD6F7E3...](#), accepted [B20699B9...](#)

Proposed fix Replace the section’s last sentence: the payee needs an `MPToken` (`MPTokenAuthorize`), otherwise `tecNO_AUTH`.

5 Vault shares carry lsfMPTCanTrade, but OfferCreate rejects them

MEDIUM documentation/tutorials · xrp1@4.6.0

`VaultCreate` sets `lsfMPTCanEscrow`, `lsfMPTCanTrade` and `lsfMPTCanTransfer` on the share issuance, with no field to change them. The [xrpl.org MPTokenIssuance](#) reference says `lsfMPTCanTrade` lets holders trade “using the XRP Ledger DEX or AMM”. `OfferCreate` accepts MPTs only under `MPTokensV2` (XLS-82), which rippled develop marks as not supported.

Repro steps

- `vault_info`: `shares.Flags 56`, i.e. `0x08, 0x10, 0x20`.
- `OfferCreate` selling shares for XRP: `temDISABLED`, a local rejection with no hash ([_probe-shares.mjs](#)).

Proposed fix State on the `MPTokenIssuance` reference that DEX trading of MPTs needs `MPTokensV2`, or leave `lsfMPTCanTrade` unset on vault shares until it ships.

6 No read command for loans or brokers

MEDIUM missing primitive · xrp1@4.6.0

A depositor’s position takes five calls, then six values computed client-side in [lib/nav.mjs](#): gross and net share value, position value, withdrawable amount, utilisation, broker debt capacity. `mpt_holders`, which lists a vault’s depositors, is documented as Clio-only.

Call	Returns
<code>vault_info</code>	totals, share issuance and flags
<code>account_objects mptoken</code>	the depositor’s shares
<code>account_objects, owner</code>	the broker, filtered on <code>VaultID</code>
<code>account_objects loan</code>	the loans, on the broker’s pseudo-account
<code>mpt_holders</code>	unknownCmd

Repro steps

- `loan_info` and `loan_broker_info` return `unknownCmd`. Neither has a documentation page.

Proposed fix Add `loan_broker_info` and `loan_info`, or list a vault's brokers and loans in `vault_info`. Return share value, withdrawable amount and utilisation. Serve `mpt_holders` from `rippled`.

7 xrpl.org omits the sole-holder exception, and zero-valued fields are absent

MEDIUM documentation/tutorials · xrp1@4.6.0

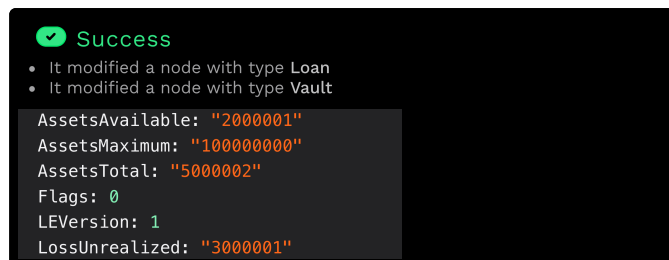
XLS-65 redeems shares at $(\text{AssetsTotal} - \text{LossUnrealized}) / \text{total shares}$, except for the sole outstanding holder, who is not charged `LossUnrealized`. The `xrpl.org` Single Asset Vault page gives the formula without the exception. `LossUnrealized` is absent until an impairment writes it, as are `AssetsAvailable`, `AssetsMaximum`, `PaymentRemaining` and `TotalValueOutstanding` at zero.

Repro steps

- 1 `vault_info` on a vault without an impaired loan: no `LossUnrealized` field.
- 2 `tfLoanImpair` on one of its loans, after `NextPaymentDueDate`.
- 3 `vault_info` returns `LossUnrealized "3000001"` with `AssetsTotal "5000002"`. Compare the explorer's *Detailed* and *Raw* tabs.

Transactions impairment [C8678C1C...](#)

Proposed fix Add the sole-holder exception to the `xrpl.org` formula. Return zero-valued fields as `"0"` in `vault_info`, or document their absence. Show changed field values in the explorer's *Detailed* tab.



Explorer. The impairment: *Detailed* tab, then *Raw* tab with the Vault node expanded.

8 One result code covers causes that need different actions

MEDIUM UX · xrp1@4.6.0

A broker who receives `tecINSUFFICIENT_FUNDS` on `LoanSet` cannot tell whether to add cover or wait for deposits. Three other codes share the problem.

Code	Cause	
tecINSUFFICIENT_FUNDS	LoanSet above the cover limit	5DC3A6E0...
	LoanSet above the vault liquidity	B92A90F5...
	LoanBrokerCoverWithdraw below the floor	4FB4A2B0...
tecLIMIT_EXCEEDED	VaultDeposit above AssetsMaximum	132BAF7C...
	AssetsMaximum set below AssetsTotal	37053F15...
	DebtMaximum set below DebtTotal	82C14ED4...
tecNO_PERMISSION	tfLoanImpair sent by the borrower	F0304194...
	tfLoanUnimpair on an unimpaired loan	13975FD3...
tecNO_AUTH	shares sent to a holder without MPToken	789AAF7E...
	non-transferable shares sent	E9DE504D...

Repro steps

- 1 `_probe-cliffs.mjs`: a `LoanSet` above the cover limit, then one above the liquidity. Both return `tecINSUFFICIENT_FUNDS`.

Proposed fix Where one transaction type maps two causes to one code (`LoanSet`, `LoanManage`, a `Payment` of shares), return distinct codes or a result message naming the check that failed.

9 No flag closes a vault to deposits

LOW missing primitive · xrp1@4.6.0

`VaultSet` exposes `Data`, `AssetsMaximum` and `DomainID`. The only way to stop deposits is a cap equal to `AssetsTotal`. `AssetsMaximum: 0` means no cap in XLS-65 and the Vault entry; the `VaultSet` reference does not repeat it.

Repro steps

- 1 `VaultSet` with `AssetsMaximum` equal to `AssetsTotal` returns `tesSUCCESS`.
- 2 A 1 XRP `VaultDeposit` returns `tecLIMIT_EXCEEDED`.

Transactions deposit refused [132BAF7C...](#)

Proposed fix Add a `VaultSet` flag that closes a vault to deposits. Repeat that `0` means no cap on the `VaultSet` reference.

Minor issues **LOW**

Issue	Category
Typos: <code>Principle0Outstanding</code> and <code>depositor</code> on the Lending Protocol page, “the the” and <code>outstanding</code> on <code>LoanManage</code> .	documentation/tutorials
<code>LoanManage</code> with no flag is an undocumented no-op (<code>tesSUCCESS</code>).	documentation/tutorials
The lending pages never say that due dates are compared with the parent ledger close time, rounded to 10 s.	documentation/tutorials
<code>validateVaultCreate</code> accepts any <code>WithdrawalPolicy</code> ; the ledger takes only 1.	client libraries
<code>validateLoanManage</code> only rejects <code>impair</code> with <code>unimpair</code> ; <code>tfLoanDefault</code> combinations reach the ledger (<code>temINVALID_FLAG</code>).	client libraries
<code>submitAndWait</code> throws on <code>tem</code> with the code only in the message text, but returns <code>tec</code> results.	client libraries

What worked

`tecHAS_OBLIGATIONS` names its cause on the three delete transactions. `VaultDelete` removes third-party `MPToken` objects and refunds their reserve. The cover floor holds to the drop, cover rates are immutable (`temINVALID`), and closed-ended phases match XLS-65.

Limits. XRP only. Intervals of 60 to 90 s. One build. No Clio, so `mpt_holders` untested. `xrpl@5.2.0` read in its source, not run. Explorer checked on `custom.xrpl.org` only. Hashes on the hackathon devnet, which may close; to replay, edit `scripts/config.mjs`. Full runs in the [README](#).